Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Fundamentals of Database Management Systems (ITUC202)

Semester: 4th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	3	4	-	7	5
Marks	100	50	-	150	

Course Description:

The Database Management Systems (DBMS) course introduces fundamental concepts and techniques for designing, implementing, and managing databases. It covers relational and NoSQL databases, ER modeling, schema refinement, and query languages like SQL and MongoDB. Advanced topics such as transaction management, concurrency control, database security, and query optimization are also explored. The course integrates practical applications with foundational theory and introduces basics of data analytics to prepare students for modern database challenges and real-world problem-solving.

Course Objectives:

1. Provide a comprehensive understanding of the relational model and relational algebra.
2. Equip students with knowledge of database design theory, including normalization and related algorithms, to ensure efficient organization of database tables.
3. Familiarize students with real-world database systems, their architectures, and components, including examples like Oracle, MySQL, MS Access, and SQL Server, MongoDB.
4. Develop proficiency in Database Management Systems, the standard language for interacting with and managing databases.

Course Outcomes:

By the end of the course, students will be able to:

1. **CO1**: Understanding of the design of database solutions to data management problems.
2. **CO2**: Effectively use the Entity Relationship Diagram for the representation of conceptual schemas.
3. **CO3**: Identify functional dependencies and apply normalization.
4. **CO4**: Understand the transaction concept and its role in transaction processing systems.
5. **CO5**: Demonstrate data retrieval procedures using the Data Manipulation Language.
6. **CO6**: Index databases and query optimization.

- **Self-study/Further Study components and materials:**

Unit No.	Unit/Topic Details	
1	**INTRODUCTORY CONCEPTS OF DATABASE MANAGEMENT SYSTEM**	
	1.1	Introduction to DBMS and its applications
	1.2	Traditional database Vs. File System
	Materials: ➢ https://www.geeksforgeeks.org/introduction-of-dbms-database-management-system-set-1/	

Syllabus:

Unit No.	Unit/Topic Details		Hours (hr)	Evaluation Weightage (%)
1	**INTRODUCTORY CONCEPTS OF DATABASE MANAGEMENT SYSTEM**		05	11
	1.1	Purpose of databases, data independence		
	1.2	Database architecture: levels, mappings, DBA roles		
	1.3	Structure of relational databases: schema, instances, integrity constraints		
	1.4	Introduction to NoSQL databases: types (document, key-value, columnar, graph), comparison with RDBMS		
	1.5	Entity-Relationship Model (Entities, attributes, relationships, and cardinality)		
	1.6	Weak entities, aggregation, generalization, specialization		
	1.7	Converting ER diagrams to relational schemas		
	1.8	Schema-less design principles in NoSQL databases		
2	**FORMAL RELATIONAL QUERY LANGUAGES**		05	11
	2.1	Relational Algebra: Selection, projection, set operations, joins		
	2.2	Tuple Relational Calculus and Domain Relational Calculus		
	2.3	NoSQL queries: find, aggregate, and pipelines in MongoDB		
3	**SCHEMA REFINEMENT AND NORMALIZATION**		12	18
	3.1	Functional dependencies, attribute closures		
	3.2	Normalization: 1NF, 2NF, 3NF, BCNF, 4NF, 5NF		
	3.3	Decomposition using Functional Dependencies		
	3.4	Schema-less design in NoSQL databases: trade-offs and use cases		
4	**TRANSACTION, CONCURRENCY AND RECOVERY MANAGEMENT**		12	18
	4.1	Transaction properties (ACID)		
	4.2	Serializability: conflict serializability and testing		
	4.3	CAP theorem and BASE properties in NoSQL		
	4.4	Locking protocols, deadlock handling, timestamp ordering		
	4.5	Log-based recovery, shadow paging, and checkpoints		
	4.6	Replication and sharding in distributed NoSQL databases (e.g., MongoDB, DynamoDB)		

5	**DATABASE SECURITY AND QUERY OPTIMIZATION**		10	15
	5.1	SQL security: roles, privileges, authorization, and views		
	5.2	Security in NoSQL: encryption, AWS DynamoDB IAM policies		
	5.3	Query cost evaluation, join strategies, query transformation		
	5.4	Aggregation pipelines and query optimization		
6	**INDEXING AND HASHING**		10	15
	6.1	Ordered indices: B+ Trees, bitmap indexing		
	6.2	Hashing: static and dynamic hashing		
	6.3	Indexing in NoSQL databases		
7	**OVERVIEW OF DATA ANALYTICS**		06	12
	7.1	Data analytics and its importance		
	7.2	Data types, data pre-processing, and exploratory data analysis		
	7.3	Integrating databases with data analytics tools (e.g., Python libraries: pandas, NumPy)		
	7.4	Data aggregation and preparation		
		Total Hours:	**60**	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	-	-	-	2	3	-	-	-	-
CO2	3	3	3	2	3	-	-	-	2	3	-	-	-	-
CO3	3	3	3	3	3	-	-	-	2	3	-	-	-	-
CO4	3	3	3	3	3	-	-	-	2	3	-	-	-	-
CO5	3	2	3	2	3	-	-	-	2	3	2	3	-	-
CO6	3	3	3	3	3	-	-	-	2	3	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:
1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High), If there is no correlation, put "-"

Practical Objectives:

- Implement SQL queries using MySQL for relational databases.
- Perform NoSQL operations using MongoDB or DynamoDB.
- Design ER diagrams and convert them to relational schemas.
- Normalize a given database schema to higher normal forms.
- Query optimization and cost evaluation.
- Explore basics of data analytics with sample datasets.

Recommended Study Material:

➢ **Text book:**

1. "Database System Concepts", by Silberschatz, Korth, and Sudarshan, 6th Edition, Publisher: McGraw Hill, ISBN: 978-0078022159.
2. "Fundamentals of Database Systems", by Elmasri and Navathe, 7th Edition, Publisher: Pearson, ISBN: 978-0133970777.

- **Reference book:**

1. "NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence", by Pramod J. Sadalage and Martin Fowler.
2. "SQL: The Complete Reference", by James R. Groff and Paul N. Weinberg.
3. "Database Management Systems", by Raghu Ramakrishnan and Johannes Gehrke, 3rd Edition.
4. "MongoDB: The Definitive Guide", by Shannon Bradshaw, Eoin Brazil, and Kristina Chodorow, 3rd Edition.

- **Additional Online Resources:**

1. MongoDB Official Documentation: https://www.mongodb.com/docs/
2. AWS DynamoDB Documentation: https://docs.aws.amazon.com/dynamodb
3. W3Schools SQL Tutorial: https://www.w3schools.com/sql/
4. Database Normalization Explained (Khan Academy): https://www.khanacademy.org/computing

- **Online Courses**

1. Data Base Management System, by Prof. P. Das and Prof. S. Chattopadhyay, IIT Kharagpuro: https://onlinecourses.nptel.ac.in/noc22_cs91/preview
2. Databases and SQL for Data Science (Coursera - IBM): https://www.coursera.org/learn/sql-data-science
3. MongoDB Basics (MongoDB University): https://university.mongodb.com/
4. SQL for Data Analysis (Udacity): https://www.udacity.com/course/sql-for-data-analysis--ud198